

- **Software Requirements Specification (SRS) – Dream Drive**

1. Introduction

1.1 Purpose

The purpose of this document is to provide a detailed description of the "Dream Drive" web application. It outlines the functional and non-functional requirements for connecting local Sri Lankan garages with car enthusiasts, focusing on vehicle restoration and specialized maintenance.

1.2 Scope

Dream Drive is a web-based platform designed to bridge the gap between garage owners (specialists in models like Lancer Box, Glanza, Evolution, etc.) and vehicle owners. The system will include a recommendation engine to match users with the best-suited service providers based on their specific needs.

1.3 Intended Users

- **Car Enthusiasts/Owners:** Individuals looking for specialized local garages.
- **Garage Owners:** Business owners looking to list their services and connect with a niche audience.
- **System Administrator:** Technical staff managing the platform and verifying listings.

1.4 Definitions & Abbreviations

- **SRS:** Software Requirements Specification
 - **FR:** Functional Requirement
 - **NFR:** Non-Functional Requirement
 - **UI/UX:** User Interface / User Experience
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2. Overall Description

2.1 Product Perspective

Dream Drive is a new, standalone web application built using Python and Streamlit. It serves as a centralized hub for the Sri Lankan automotive subculture, moving away from fragmented social media groups (TikTok/Facebook) into a structured service marketplace.

2.2 User Classes

1. **Customer (Guest/Registered):** Can search and view garage recommendations.
2. **Garage Partner:** Can create and manage garage profiles and service lists.
3. **Admin:** Oversees the database, moderates reviews, and manages the recommendation logic.

2.3 Operating Environment

- **Platform:** Web-based.
- **Frontend/Backend:** Streamlit (Python).
- **Data Storage:** Pickle files (.pkl) / CSV / Database for garage lists and artifacts.

2.4 Assumptions & Constraints

- Users have stable internet access.
 - The recommendation system relies on the accuracy of the `garage_list.pkl` data.
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3. Functional Requirements (FR)

ID Requirement Description

FR1 The system shall allow **Garage Owners** to register and create a profile listing their specialties (e.g., Engine tuning, Bodywork, Restoration).

FR2 The system shall allow **Customers** to search for garages by location, vehicle model (e.g., Lancer Box), or service type.

FR3 The system shall provide a **Recommendation Engine** that suggests the top 5 garages based on user input and similarity scores.

FR4 The system shall allow users to view garage contact details and location via a "one-click" connection button.

FR5 The system shall display a dashboard for the **Admin** to update the garage database.

4. Non-Functional Requirements (NFR)

- **Performance:** The recommendation engine shall process user queries and display results in under 3 seconds.
 - **Usability:** The UI shall be optimized for both desktop and mobile browsers since many car enthusiasts use TikTok/Instagram on mobile.
 - **Security:** Access to the artifacts folder and admin features shall be restricted to authorized personnel.
 - **Reliability:** The system shall maintain 99% uptime during peak hours (Sri Lanka Standard Time).
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5. Use Case Diagram (Summary)

- **Actor: Customer** -> Use Cases: [Search Garage], [View Recommendations], [Contact Garage].
 - **Actor: Garage Owner** -> Use Cases: [Register Business], [Update Service List].
 - **Actor: Admin** -> Use Cases: [Manage Database], [Monitor System Performance].
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Evidence of Requirement Elicitation (Appendix A)

Stakeholder Discussion Summary: To gather these requirements, interviews were conducted with local car enthusiasts in Pannipitiya and Rukmale.

- **Key Insight:** Users expressed frustration that finding a specialist for "Evolution" or "BJ45" restoration requires asking around on social media; they need a one-click directory.
 - **Evidence:** *[Insert a screenshot here of your survey results or a photo of your interview notes regarding garage needs].*
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Requirement Analysis Method

Requirement Prioritization (MoSCoW):

- **Must Have:** Garage search, recommendation logic, "One-click" contact.
- **Should Have:** User reviews, photo gallery for restorations.
- **Could Have:** Integration with Google Maps for navigation.